ADRIÀ MITJAVILA VENTURA

Bioinformatician | LaCaixa INPhINIT Fellow | PhD student at *Institut Josep Carreras* | MSc

Bioinformatician with training in Molecular Genetics and Biotechnology, and more than 5 years of experience working in well known research centers in Spain and Italy.

With a solid background in Genomics and Transcriptomics, I aim to analyse large NGS data sets and develop analysis workflows to solve complex biological problems in the context of evolution, gene regulation and cancer.

Innovative and agile thinker always striving to face a challenge and find the right solution. I enjoy working independently as well as in a collaborative environment to deliver the best results. With experience in mentoring and support students, and in scientific training and outreach activities.

I look forward to joining an international team in the healthcare sector where I can make the most of my skills and to grow in a stimulating work environment.

EXPERIENCE

11/2022 | Present — **PhD student**
Institut Germans Trias i Pujol & *Institut Josep Carreras* — Badalona, Spain
Group of Regulatory Genomics (Dr. Tanya Vavouri)

08/2021 | 11/2022 — **Bioinformatician & master's thesis student**
Institut Josep Carreras — Badalona, Spain
Group of Regulatory Genomics (Dr. Tanya Vavouri)

07/2019 | 07/2021 — **Bioinformatician**
Istituto Europeo di Oncologia — Milan, Italy
Group of Epigenetic Mechanisms in Cancer (Dr. Diego Pasini)

09/2017 | 10/2018 — **Master's thesis student & Research assistant**
Instituto de Biomedicina de Sevilla — Seville, Spain
Group of Gene Expression (Dr. Sebastián Chávez)

09/2016 | 06/2017 — **Internship & bachelor's thesis student**
Microbial. Sistemes i aplicacions analítiques, S.L. — Girona, Spain

EDUCATION

2022 | Present — **PhD in Bioinformatics**
Universitat Autònoma de Barcelona (UAB) — Barcelona, Spain

2018 | 2022 — **MSc in Bioinformatics and Biostatistics**
Universitat Oberta de Catalunya (UOC) — Online

- Master's thesis: *Analysis of the variation in piRNA expression across three close related Mus species*. Grade: 9.7/10 (*with honours*). Supervisor: Dr. Tanya Vavouri.

2017 | 2018 — **MSc in Molecular Genetics and Biotechnology**
Universidad de Sevilla (US) — Seville, Spain

- Master's thesis: *Influence of Xrn1 in nucleosome positioning across Saccharomyces cerevisiae genome and its effects in transcription*. Grade: 9.2/10. Supervisors: Dr. Sebastián Chávez de Diego & Dr. Gabriel Gutiérrez Pozo.

2012 | 2017 — **BSc in Biotechnology**
Universitat de Girona (UdG) — Girona, Spain

- Bachelor's thesis: *Butyrate-induced changes in the diversity of intestinal mucosa-associated microbiota in colorectal cancer patients submitted to a lateral ileostomy*. Grade: 9.4/10. Supervisor: Prof. Jesús García-Gil.

Girona (Spain)
adria.mitjavila94@gmail.com
amitjavilaventura.github.io
/adria-mitjavila-ventura
/amitjavilaventura
0000-0002-0337-1929

SKILLS

Informatics

R/Bioconductor, Bash, AWK, Python (basic)
Snakemake, Conda, Docker, Singularity, HPC (PBS & SLURM)
Linux (Ubuntu), Windows, MacOS

Bioinformatics

(small) RNA-/ChIP-/ATAC-/MNase-seq data analysis
Single-cell RNA data analysis
Spatial transcriptomics (10x Xenium) custom panel design and data analysis
UCSC, ENSEMBL and IGV genome browsers
Multiple sequence alignment, ENSEMBL Compara, BLAST, QIIME2

Languages

Catalan (Native)
Spanish (Bilingual)
English (IELTS 7.5 - C1)
Italian (Basic)

DEVELOPMENT

`ggmitji`: complements and customization for ggplots.

ATAC-seq Snakemake pipeline

More on my GitHub profile.

HONORS & AWARDS

INPhINIT Incoming PhD Fellowship (LaCaixa, 2022-2025)

Master's thesis, *with honours* (UOC, 2022)

SEMM PhD Fellowship (2019-*declined*)

Elite athlete (BOE, 2015-2021)

World Rowing U23 Champion (FISA, 2016)

References available upon request.
Build with pagedown, code here.
Last updated on 2026-05-28.

FURTHER TRAINING

2023
2024
2025

INPhINIT PhD Fellowship annual gathering
La Caixa

- Three 4-day annual gatherings with other PhD fellows in which we worked on soft skills, networking, career development, wellbeing and other topics related to academy and industry.

2025

R for multiomics data analysis
Decode Life

- R data analysis, Transcriptomics, Metabolomics

03/2023

International course on Non-codiong genome
Institute Curie

- Poster presentation: *Fast evolution of piRNA cluster expression in mouse*.
- Evaluation of MSc students in a Journal Club.

PUBLICATIONS

2026

Mitjavila-Ventura & Vavouri. Mechanisms seeding the emergence of new piRNAs.
Trends in Genetics. 2026. doi: 10.1016/j.tig.2026.04.004

2025

Casas, Mitjavila-Ventura *et al.* Genetic polymorphisms lead to major, locus-specific, variation in piRNA production in mouse.
EMBO Journal. 2025. doi: 10.1038/s44318-025-00475-4

2021

Begley *et al.* Xrn1 influence on gene transcription results from the combination of general effects on elongating RNA pol II and gene-specific chromatin configuration.
RNA Biology. 2021. doi: 10.1080/15476286.2020.1845504